A Privacy-Preserving, Autonomous Offensive Security Agent using Edge AI and Visual Reasoning

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Abstract:

This project uses the Red Agent part of the ELISAR (Extended Large Language Model of Information Security and Risk) framework, which is made to run on consumer grade hardware (Edge AI). The proposed goal is to create an independent offensive security assistant, which would use Retrieval-Augmented Generation (RAG) to emulate adversarial behaviors and detecting such security lapses at their initial stages, before they are exploited.

The system architecture is fully implemented on a standard laptop (16GB RAM) on the ELISARCyberAIEdge7B model and DeepSeek-R1 to reason, organized through Ollama and LangGraph. The agent loads the MITRE ATT&CK knowledge base into a local vector database (ChromaDB) where it is able to match the characteristics of observed systems with known adversary tactics.

One of the novelties of such implementation is the implementation of the Automated Attack Graph Visualization, in comparison with the old text-based chatbot, this Red Agent works with a Chain-of-Thought planning module, which produces the structured diagrams in the form of mermaid.js. This enables the agent to not only explain an attack plan, but to visually plot the logical flow of Initial Access to Impact, aiding the explainability of the plan greatly. The end product will be a workable, air-gapped prototype that has the ability to accept target parameters and generate a holistic, visualized attack strategy that proves that advanced red-teaming can be democratized and placed at the edge.

Keywords: Edge AI, Red Teaming, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Autonomous Security Agents, MITRE ATT&CK, Attack Graph Visualization, Explainable AI (XAI), Cybersecurity, On-device Inference.

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Project Guide

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